RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch	**:** V Semester B.E CSE
Course Code & Title	**:** CCS366 & Software Testing and Automation
Name of the Faculty member	**:** Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic: Unit V/Test Automation and Tools/**Different Web Drivers
- **Course Outcome:5**
- **Topic Learning Outcome: 5b**
- **Activity Chosen:** Collaborative Learning
- **Justification:**
- Students can work together to understand how each web driver functions, such as ChromeDriver, GeckoDriver, and EdgeDriver, making it easier to compare their behavior in automation. Collaborative learning helps students quickly solve configuration and version issues, as they can support each other in troubleshooting common driver errors.

- Working in teams allows students to share different approaches and scripts, helping everyone learn multiple ways of using web drivers effectively. This method builds teamwork and communication skills, which are essential for real-world software testing projects involving cross-browser automation..
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
- The collaborative learning activity on different web drivers was implemented step by step to help students gain hands-on experience in Selenium automation. The session began with an introduction to various web drivers such as ChromeDriver, GeckoDriver, and EdgeDriver, ensuring that all students understood their purpose and role in browser automation.
- Pre-class materials were shared so that learners could review installation steps, browser compatibility, and sample Selenium scripts beforehand. During the activity, students were divided into small groups, and Ms. Lakshitha and her team were assigned the ChromeDriver to explore. They downloaded, configured, and tested the driver by running basic scripts such as launching a webpage and performing simple actions. As they worked together, students encountered common issues like path errors and version mismatches, which they collaboratively solved, strengthening their troubleshooting skills. Figure 1 shows that Presentation and idea sharing by Lakshitha's team. Figure 2 shows that Presentation and idea sharing by Lakshitha's team.
- After completing the configuration, each team executed sample scripts and analyzed the behavior of their assigned web driver. Lakshitha's team observed the speed and reliability

of ChromeDriver, while other teams shared insights on GeckoDriver and EdgeDriver. The groups then discussed their findings, comparing performance, compatibility, and challenges. Selected teams, including Lakshitha's, presented their results to the class, explaining the working, advantages, and technical issues they faced.

- The instructor concluded the session by clarifying doubts, summarizing key differences among drivers, and emphasizing the importance of cross-browser testing in automation. Students later reflected on the activity, with Lakshitha's team expressing that the collaborative approach and hands-on practice significantly enhanced their confidence and understanding of automation using multiple web drivers.

- **CO – PO / PSO mapping:**

CO	**PO1**	**PO2**	**PO5**	**PO8**	**PO10**	**PO12**	**PSO3**
CO4	**3**	**3**	**3**	**3**	**3**	**3**	**3**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO4**	**PO5**	**PO10**	**PO12**	**PSO2**
	3	**3**	**3**	**1**	**2**	**2**	**2**
Justification for correlation	Students able to applying automation techniques with Selenium and TestNG requires a solid understanding of engineering principles to develop robust	Students able to utilize Selenium and TestNG for automated testing involves analyzing testing requirements and identifying suitable automation strategies to address them effectively.	Students able to Automate software testing with Selenium and TestNG involves investigating complex testing scenarios and devising automated testing approaches to address them.	Students able to Leverage Selenium and TestNG for automated testing demonstrates proficiency in utilizing modern testing tools and technologies to enhance testing efficiency and effectiveness.	Students able to effective communication of automated testing strategies and results to stakeholders and team members ensures alignment and support for automated testing initiatives.	Students able to applying Selenium and TestNG for automated testing necessitates a commitment to continuous learning to stay updated with evolving automation techniques and best practices.	Students able to automate software testing with Selenium and TestNG aligns with the application of modern tools and technologies, contributing to the reliability of IT solutions in IoT, cloud computing, and cybersecurity domains.

- **Images / Screenshot of the practice:**

Figure 1: Presentation and idea sharing by Lakshitha’s team.

Figure 2: Presentation and idea sharing by Padma Pooja’s team.

Reflective Critique:

❖ **Feedback from Students and Stakeholders:**

- Students shared that the collaborative activity helped them understand the differences between ChromeDriver, GeckoDriver, and EdgeDriver more clearly through hands-on experience.
- Stakeholders appreciated that students worked in teams to troubleshoot real configuration issues such as driver version mismatches and path errors, which strengthened practical learning.
- Many students felt that discussing problems within teams, including Lakshitha's team, increased their confidence in handling cross-browser automation.

❖ **Benefits of the Practice:**

- Collaborative learning enabled students to share knowledge, compare driver behavior, and understand cross-browser testing in a practical manner.
- Teamwork improved problem-solving skills, as students could quickly identify and fix driver errors with peer support.
- Active participation made the learning process more engaging, helping students build confidence in Selenium automation tools.
- Students gained real-time experience in configuring and executing different web drivers, which is essential for industry-level automation projects.

❖ **Challenges Faced:**

- Some students initially struggled with technical issues such as incorrect driver versions and system path configurations, requiring additional support.
- Slow learners found it challenging to keep pace with hands-on troubleshooting but were eventually supported by their teams.
- A few students hesitated to communicate or present their findings to the entire class, affecting overall participation.
- Differences in system setups (browser versions, OS compatibility) caused delays for some teams during the configuration stage.

References:

❖ https://en.wikipedia.org/wiki/Peer_learning
❖ Peer to Peer Learning - Examples, Benefits & Strategies
❖ https://whatfix.com/blog/peer-to-peer-learning/

Signature of Faculty Member **HOD**